

Characterizing Human Skin Mechanical Properties with a Uniaxial Extensometer

Justin Huang, Katie Latvakoski, Nadia Qazi, Talia Green, and Nancy Lape*
Department of Engineering, Harvey Mudd College, Claremont, California

Background & Motivation

Skin mechanical properties tested *in vivo* are critical for cosmetics [1], scar management [2], and designing biomimetic materials [3] that replicate skin's behavior. Mechanical properties have been determined *in vivo* for plane shear, indentation, suction, or torsion to stretch the skin [1,4-5]; however, few devices utilize *in vivo* uniaxial stretching. Moreover, mechanical extension, such as uniaxial stretching, has been shown to alter skin permeability, influencing how substances penetrate or are retained by the epidermal barrier [6]. Although previous work exists with *in vitro* experiments with uniaxial stretching, they do not holistically replicate interactions with all sublayers of skin or the natural furrows in the SC *in vivo*. Furthermore, because *in vitro* studies are usually done with dried and flattened skin, the failure strain is notably different from *in vivo* experiments.

In order to test skin mechanical properties, a lab-built *in vivo* uniaxial extensometer (George), starts with preconditioning in order to yield consistent skin characterizations and then performs the creep and stress relaxation tests to characterize a participant's skin. Creep holds a participant's skin at a constant force and measures changes in displacement, while stress relaxation holds the skin at a constant strain and measures the force. Fractional viscoelastic models use springpots to characterize the viscoelastic power-law data with fewer parameters than standard spring and damper models [7]. The skin's mechanical properties were characterized by fitting creep and stress relaxation data using the fractional Kelvin-Voigt (FKV) and fractional Poynting-Thomas (FPT) models shown in Figures 1 and 2.

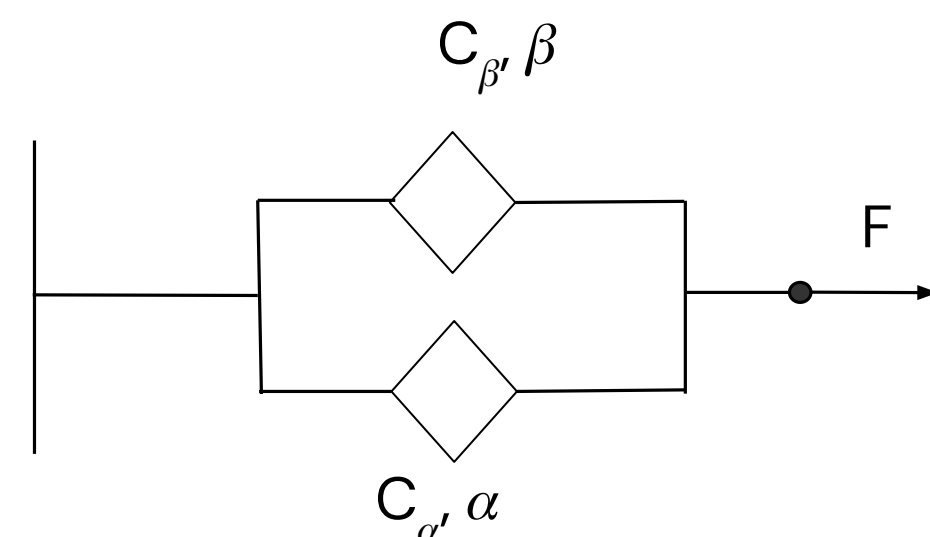


Figure 1. FKV Model

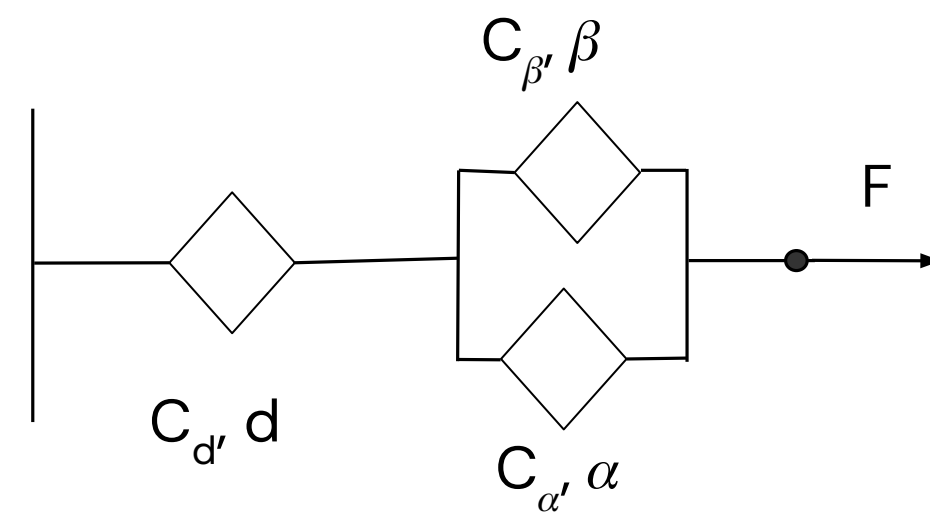


Figure 2. FPT Model

Mechanical Design & Updates

George is a lab-built uniaxial extensometer that stretches the skin *in vivo* on the ventral forearm.

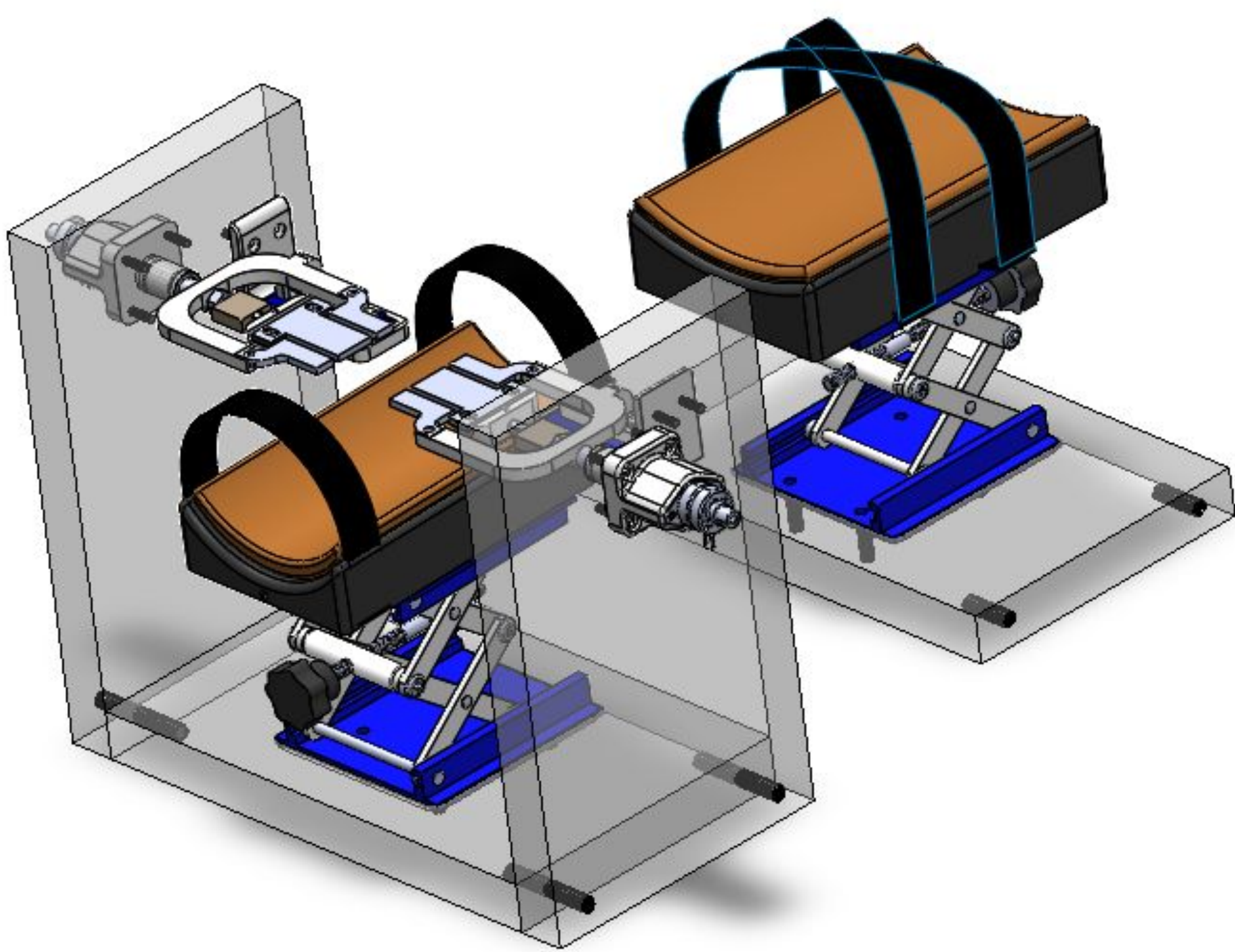


Figure 3. SOLIDWORKS model of George displaying the dual armrest setup

This summer, different components were added or updated to George with the goal of improving comfort and reliability for George tests.

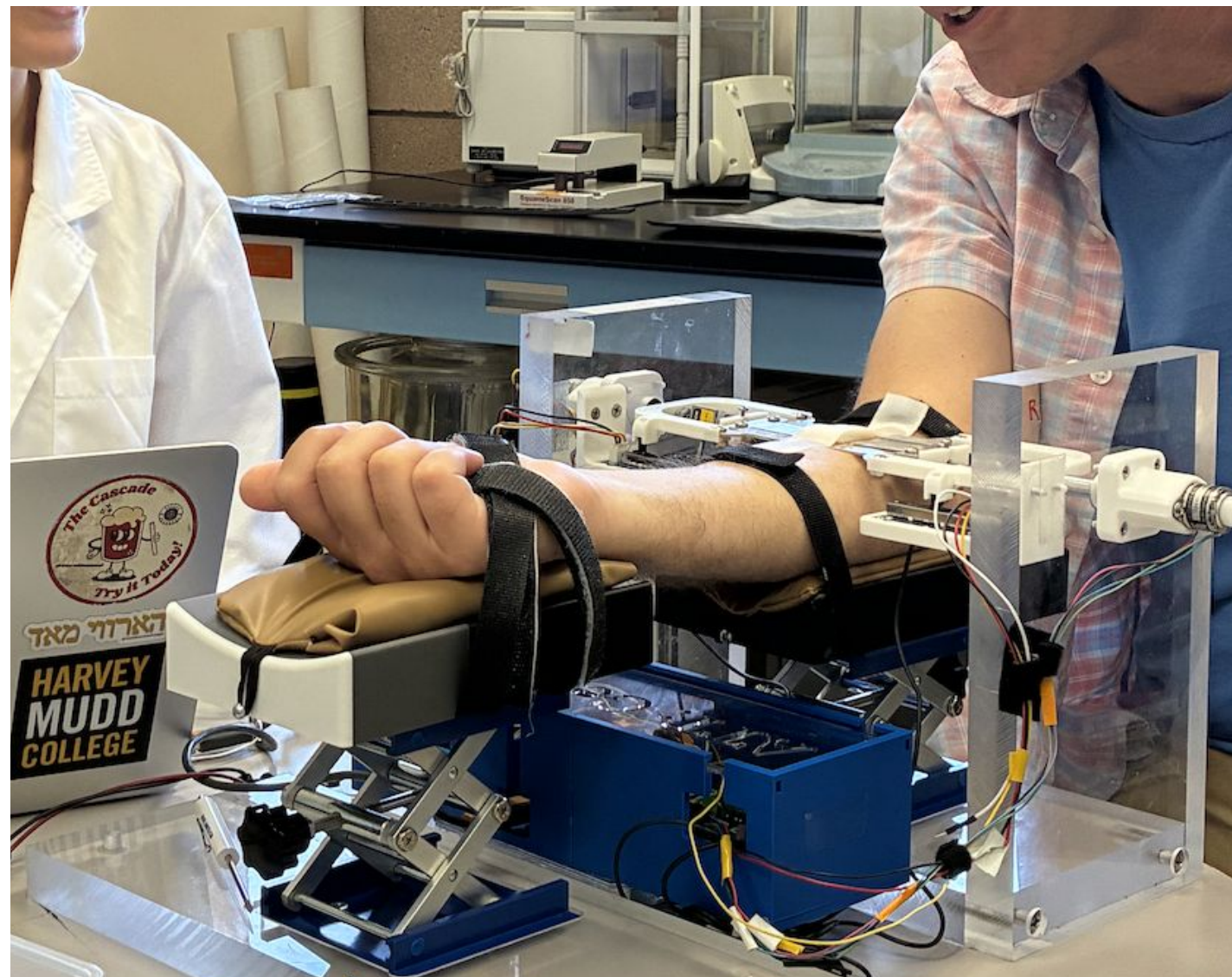


Figure 3. Dual Armrests to Improve Testing Stability and Comfort

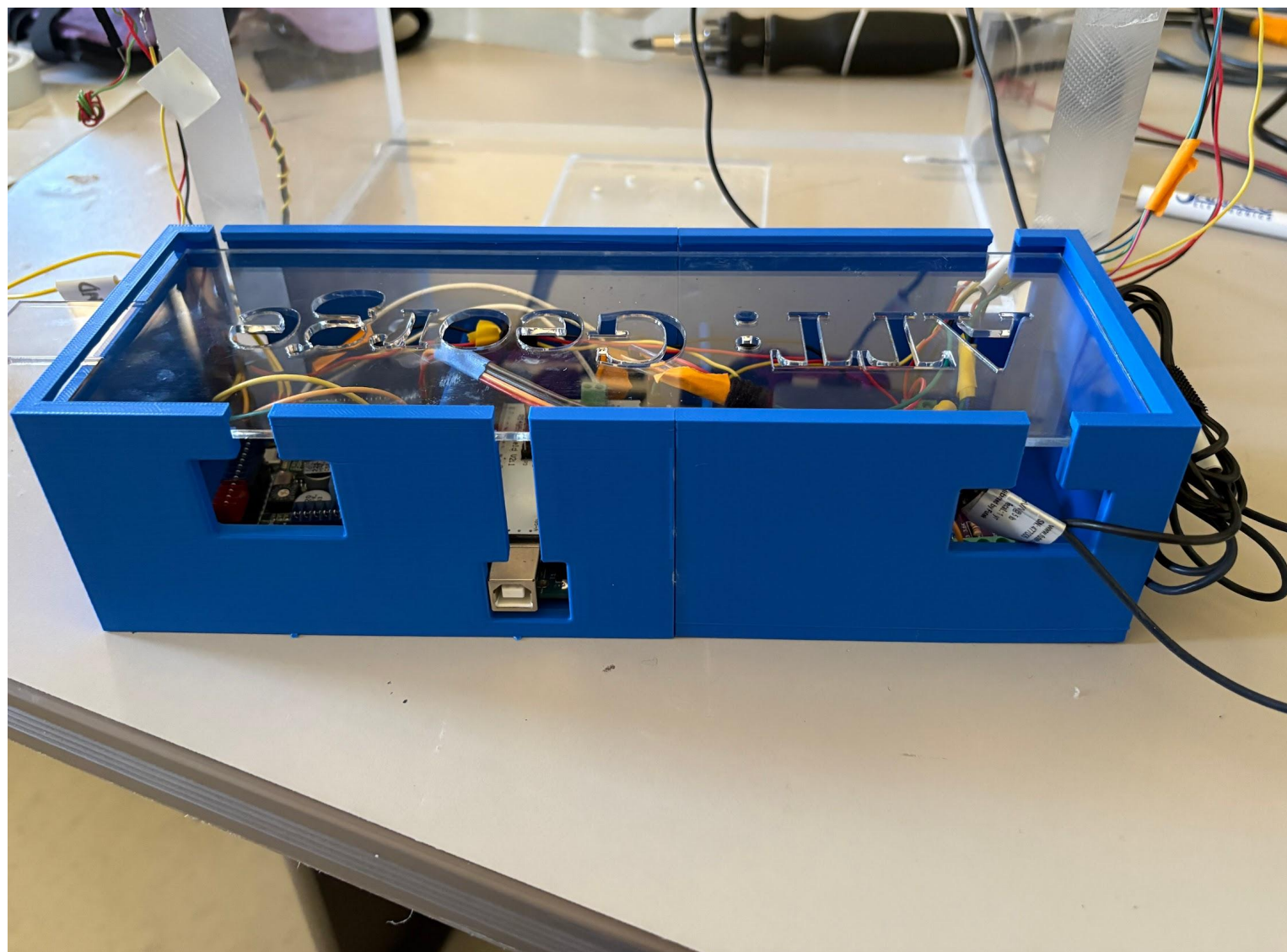


Figure 4. Cable Box to Store George Electrical Components

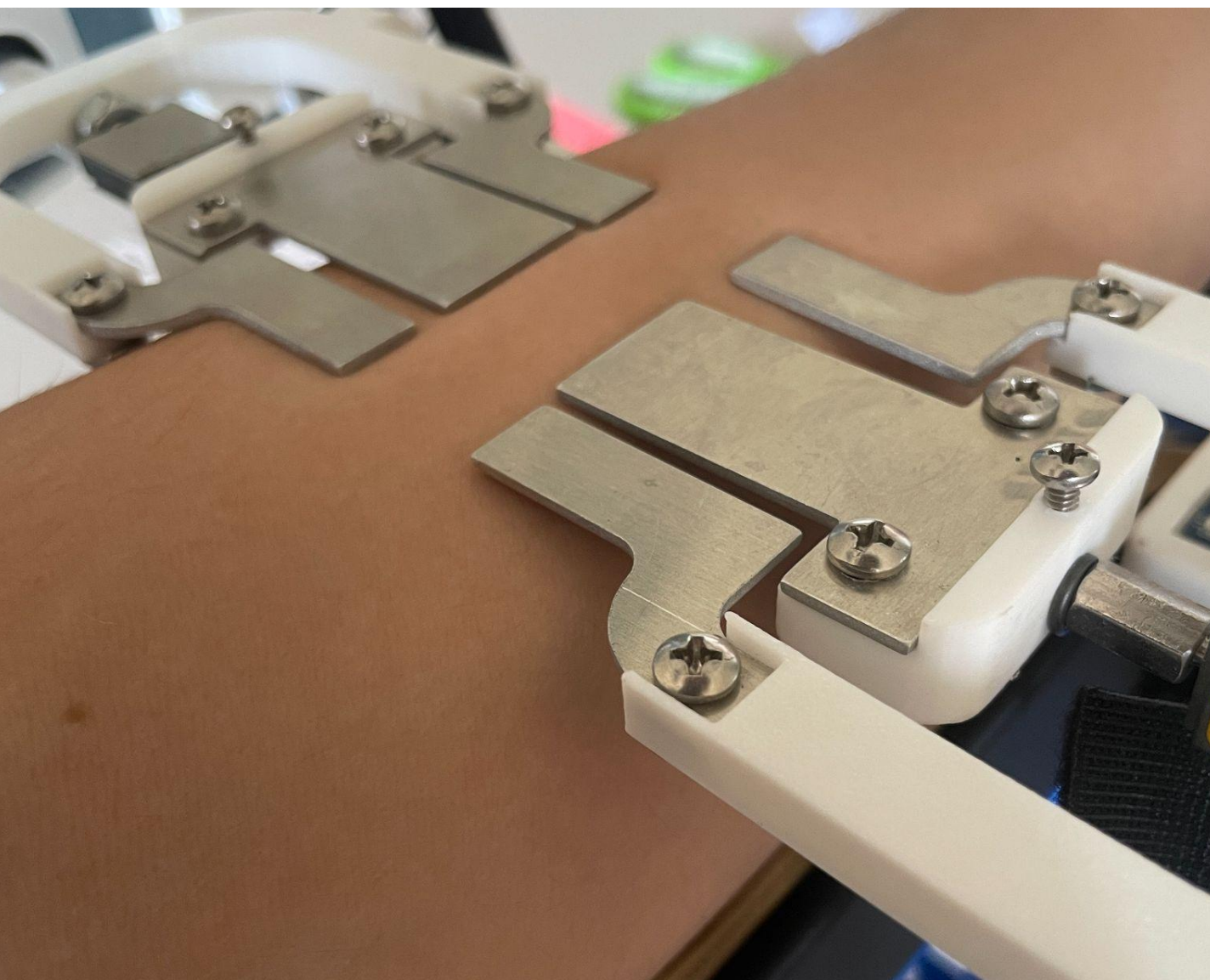


Figure 5. Aluminum Tabs to Provide More Consistent Test Results

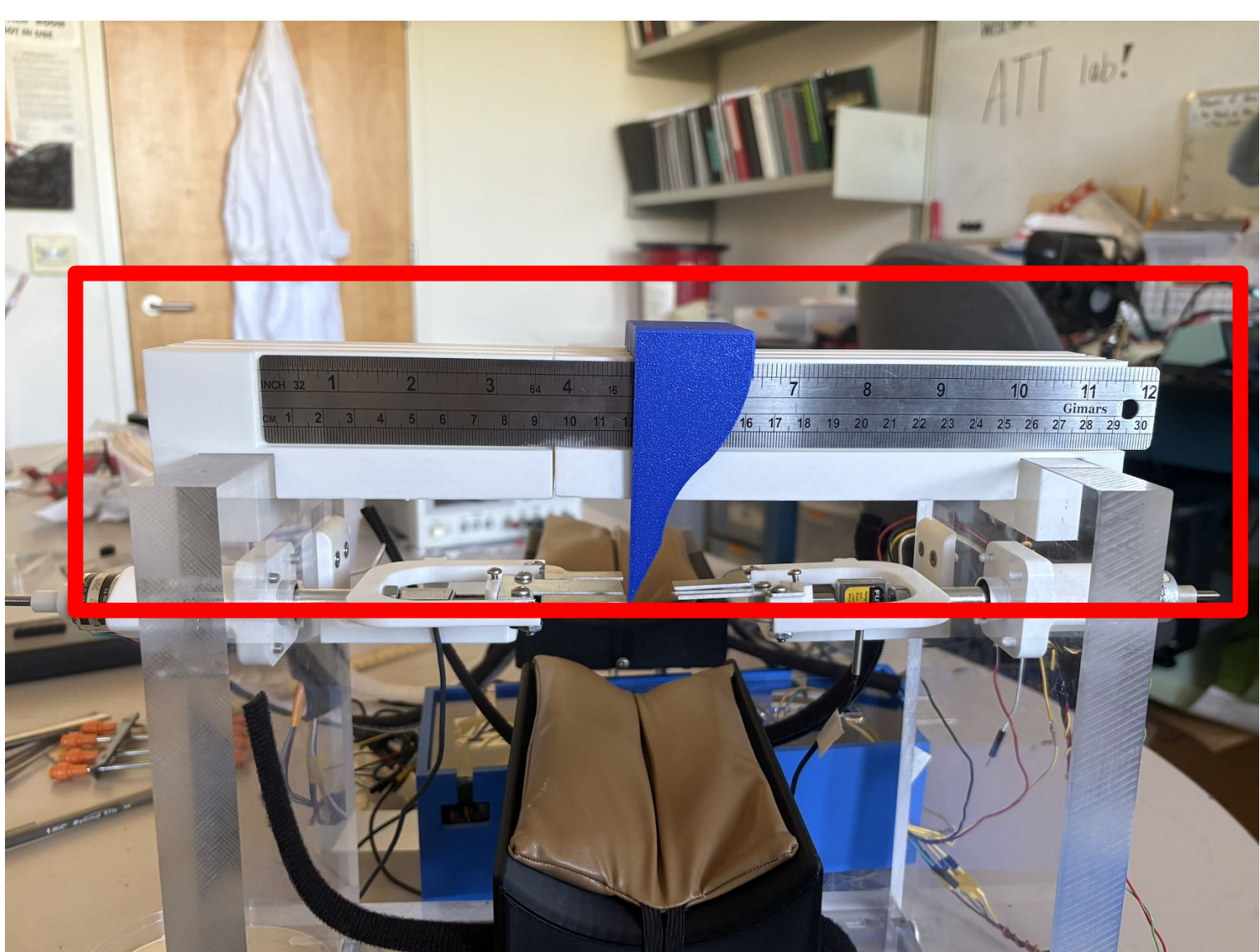


Figure 6. Calibration Slider Tool to Simplify Tab Calibration

Methods

George **tests the mechanical properties of skin** using the following tests:

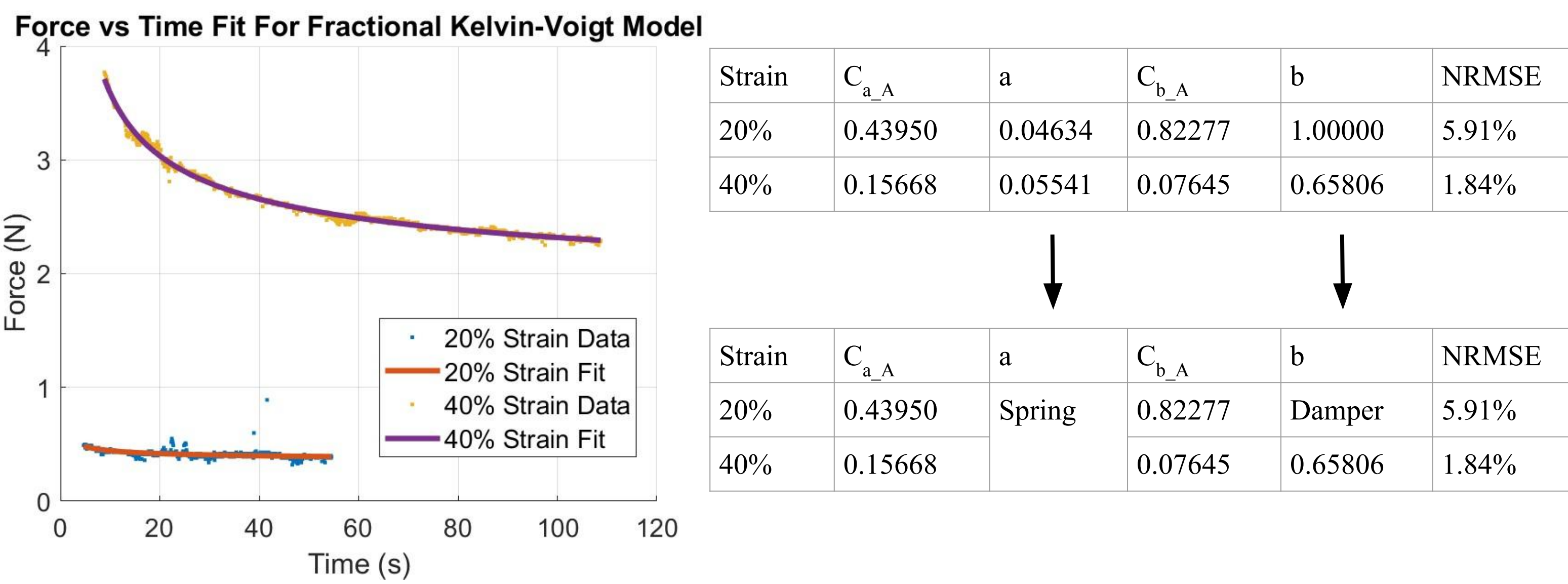
- Creep: constant force on the skin is maintained
 - Displacement actively increases over time in viscoelastic materials
 - Data: **strain (%)** of the skin **vs time**.
- Stress Relaxation: constant displacement on skin is maintained
 - Force actively decreases over time for viscoelastic materials
 - Data: **force** exerted on the tab by the skin **vs time**.

Data processing done in MATLAB:

- Fractional viscoelastic models were fit to the stress relaxation and creep data.
 - Models were evaluated on NRMSEs and visual fits and the best two were chosen.

Experimental Data and Results

Experimental data captures the characteristic viscoelastic behavior of skin in the creep and stress relaxation tests. The skin's mechanical properties are best extracted via the FKV model for stress relaxation and the FPT model for creep.



Figures 7a. Stress relaxation data fit with FKV model and 7b. FKV fit parameters where area is combined with the springpot coefficients

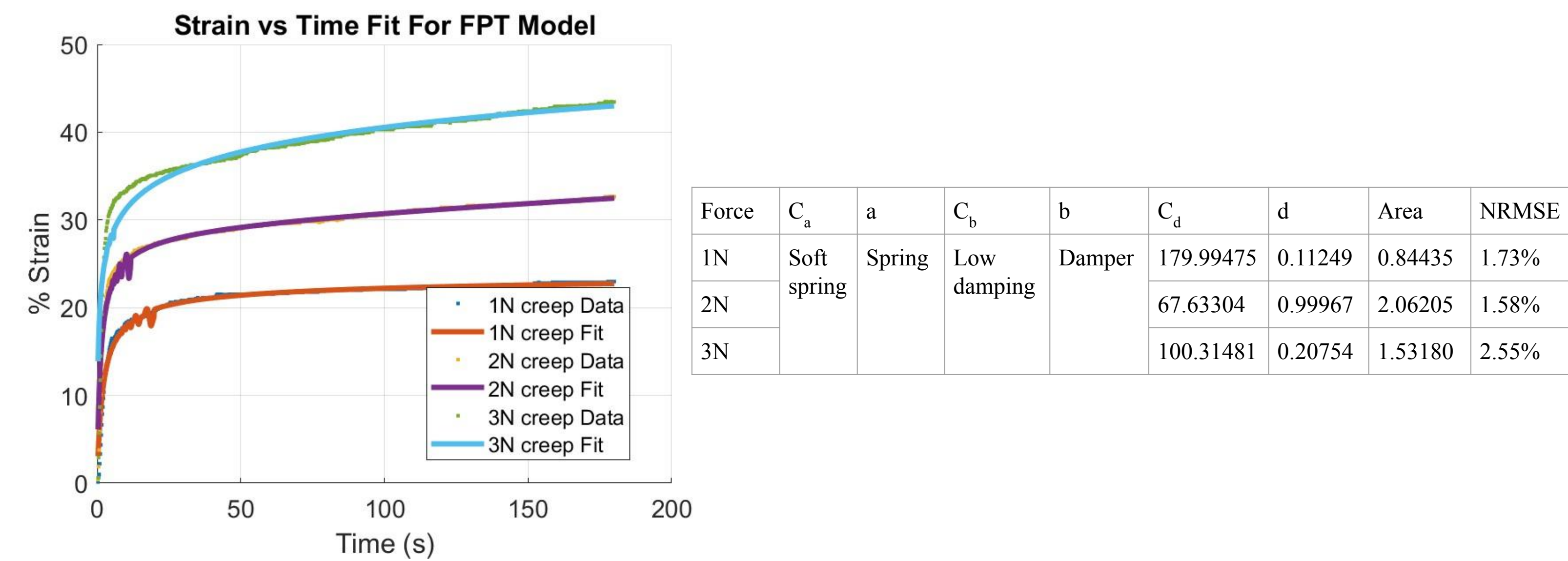


Figure 8a. Creep data fit with FPT and 8b. FPT fit parameters where area isn't combined with the springpot coefficients

The fractional Kelvin-Voigt model fits the stress relax data well with few parameters which allows for easier interpretability of the participant's data. This will allow comparisons between and within participant tests.

The FPT model captures the initial jump in strain for the creep data and the linear increase after fifty seconds. However, the model does introduce ringing between zero and twenty-five seconds, which is not characteristic of the data itself. Further research into the model is necessary to discover why this occurs.

Conclusions, Future Work, and Impact

George is able to capture mechanical property testing data, which can be analyzed with fractional viscoelastic models: specifically the FKV model can be used to capture stress relaxation data and FPT model can be used to capture creep data fairly well. However, the ringing in the FPT model isn't characterizing the data, which means that further models or the fitting code itself should be analyzed.

Additionally, because George doesn't apply force to a uniform cross section of area, and rather at the top of the skin, there is no current way of measuring the area that is affected by the force on the skin. In other models, this area parameter can be combined with the springpot coefficients, but this isn't possible with the FPT model. Thus, the FPT model can fit an extra parameter that should be fixed for a specific force and person. Finding an approximate area in order to verify or fix the area parameters in a range may be possible through ultrasound [8] or modeling [9].

These results demonstrate the ability of fractional viscoelastic models to model skin and meaningfully describe the skin. The stress relaxation data in conjunction with the FKV model can be used to compare mechanical properties across age groups, between participants, and within participants themselves, which the team intends to do in the fall.

References and Acknowledgements

References:

- Khatyr, F. (2004). Model of the viscoelastic behaviour of skin *in vivo* and study of Anisotropy. *Skin Research and Technology*, 10(2), 96–103.
- Blair MJ (2020). Skin Structure-Function Relationships and the Wound Healing Response to Intrinsic Aging. *Adv Wound Care* (New Rochelle). 2020 Mar 1;9(3):127-143
- Shi, S. (2023). A skin-like nanostructured membrane for advanced wound dressing. *Composites Part B: Engineering*, 250, 110438.
- Lévêque, J. L. (1988). Are corneocytes elastic? *Dermatology*, 176(2), 65–69.
- Pedersen, L. (2006). Mechanical properties and barrier function of healthy human skin. *Acta Dermato-Venereologica*, 86(4), 308–311.
- Vyumvuhore, R. (2013). Raman spectroscopy: a tool for biomechanical characterization of Stratum Corneum. *Journal of Raman Spectroscopy*, 1077-1083.
- Bonfanti, A., L. Kaplan, J., Charras, G., & Kabla, A. (2020). Fractional viscoelastic models for Power-Law materials. *Soft Matter*, 16(26), 6002–6020.
- Manschot, J. F. M., & Brakkee, A. J. M. (1986). The measurement and modelling of the mechanical properties of human skin *in vivo*—I. the measurement. *Journal of Biomechanics*, 19(7), 511–515.
- Yang, G., Pang, X., Liu, A., Li, C., Zhu, J., Huang, R., & Ma, X. (2024). Design and validation of a DMA-inspired device for In-Vivo measurement of Human skin mechanics: A finite element analysis approach. *Journal of Physics: Conference Series*, 2851(1), 012019.

Acknowledgements:

This work is funded by The Patton and Claire Lewis Fellowship in Engineering Professional Practice, the Norman F. Sprague III, M.D. Experiential Learning Fund, and the Rose Hills Foundation.